Descriptive Statistics Project.

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Project:

Descriptive statistics of the average precipitation (in inches) for Miami Area, FL from January, 2024 to December, 2024.

URL: https://www.weather.gov/wrh/Climate?wfo=mfl

Navigation:

Location: Miami Area, FL

Product: Monthly Summarized Data

Year Range: January, 2024 - December, 2024

Variable: Precipitation (in inches)   Summary: Sum

Description: The monthly total precipitation (in inches) for Miami Area, FL from January, 2024 to December, 2024.

$0.89, 3.03, 4.78, 2.78, 4.21, 17.34, 5.44, 12.83, 9.11, 7.5, 0.47, 1.18$

NOWData - NOAA Online Weather Data

Enlarge results | Print

Monthly Total Precipitation for Miami Area, FL (ThreadEx)

Click column heading to sort ascending, click again to sort descending.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2015	0.95	2.81	1.32	4.03	2.39	3.60	5.91	9.02	9.97	4.70	7.55	9.82	62.07
2016	7.57	2.85	0.61	1.09	8.27	8.72	4.11	13.77	6.05	9.62	0.94	2.33	65.93
2017	3.40	1.31	3.92	2.16	2.69	15.97	12.45	8.57	14.97	12.61	4.41	0.66	83.12
2018	1.01	0.37	0.19	3.80	16.59	7.89	8.02	9.58	7.89	1.76	2.77	1.58	61.45
2019	2.00	1.53	1.91	3.16	4.61	12.43	10.54	15.74	3.25	3.93	1.62	6.43	67.15
2020	2.03	2.90	0.10	3.86	18.89	7.11	10.26	7.44	10.92	11.86	9.60	1.60	86.57
2021	0.50	2.90	1.87	2.81	2.68	12.44	8.18	7.29	12.12	4.91	4.38	1.19	61.27
2022	5.98	1.60	2.03	4.66	4.47	15.61	4.81	5.17	12.15	4.85	8.49	1.73	71.55
2023	0.05	5.72	4.50	9.95	5.92	7.87	7.42	10.25	12.76	5.78	9.36	3.85	83.43
2024	0.89	3.03	4.78	2.78	4.21	17.34	5.44	12.83	9.11	7.50	0.47	1.18	69.56
2025	0.89	0.72	1.59	1.03	7.98	8.41	8.92	5.88	16.55	3.48	M	M	M
Mean	2.30	2.34	2.07	3.58	7.15	10.67	7.82	9.59	10.52	6.45	4.96	3.04	71.21
Max	7.57 2016	5.72 2023	4.78 2024	9.95 2023	18.89 2020	17.34 2024	12.45 2017	15.74 2019	16.55 2025	12.61 2017	9.60 2020	9.82 2015	86.57 2020
Min	0.05 2023	0.37 2018	0.10 2020	1.03 2025	2.39 2015	3.60 2015	4.11 2016	5.17 2022	3.25 2019	1.76 2018	0.47 2024	0.66 2017	61.27 2021

Objectives:

I shall do the following objectives.

(1.) Dataset Description: Describe the dataset.

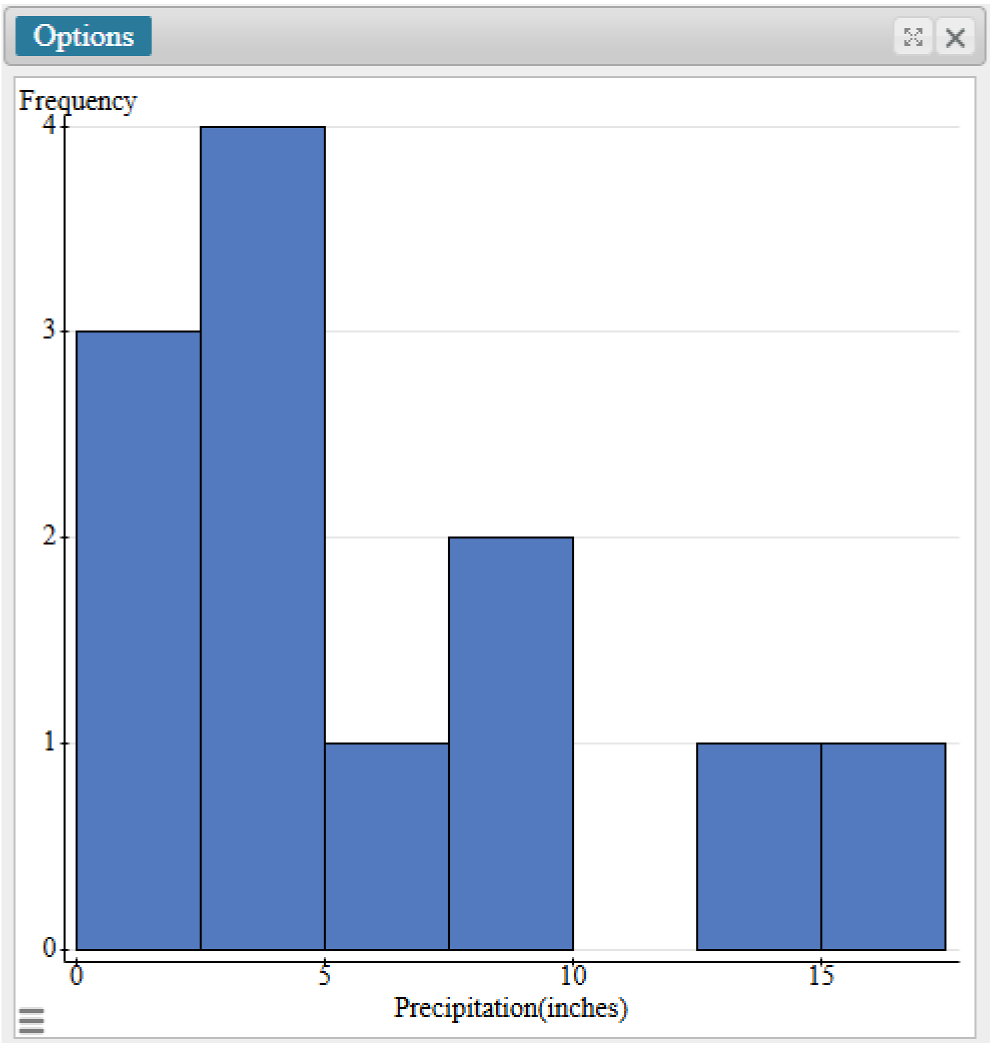
(2.) Dataset presentation: Present the dataset using a histogram.

(3.) Measures of Central Tendency: Calculate and interpret the measures of central tendency.

(4.) Measures of Variability: Calculate and interpret the measures of variability.

(5.) Measures of Location: Calculate and interpret measures of location.

Data Presentation:



[2.5, 5)

Four months had precipitation between 2.5 inches (included) and 5 inches (excluded).

[12.5, 15)

One month had precipitation between 12.5 inches (included) and 15 inches (excluded)

[15, 17.5)

One month had precipitation between 15 inches (included) and 17.5 inches (excluded)

Data Analysis.

We shall begin with measures of center. I shall round all final answers to two decimal places as needed.

Options

Summary statistics:

Column	n	Sum	Mean	Median	Mode	Min	Max	Range	Variance	Std. dev.	Q1	Q3	IQR
Precipitation(inches)	12	69.56	5.7966667	4.495	No mode	0.47	17.34	16.87	26.43557	5.1415532	1.98	8.305	6.325

We shall use formulas, Pearson Starcrunch software, and Texas Instrument (TI-84 Plus) Calculator.

Measures of Center:

These are Mean, Median, Mode, and Midrange.

$0.89, 3.03, 4.78, 2.78, 4.21, 17.34, 5.44, 12.83, 9.11, 7.5, 0.47, 1.18$

$Sample\ size = 12$

Mean:

$$Mean = \frac{Sum\ of\ Data\ Values}{Sample\ Size} = \frac{69.56}{12} = 5.7966667 \cong 5.80$$

The average monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is approximately 5.80 inches.

Median: I shall average the dataset in ascending order.

$0.47, 0.89, 1.18, 2.78, 3.03, 4.21, 4.78, 5.44, 7.5, 9.11, 12.83, 17.34$

$$Median = \frac{4.21 + 4.78}{2} = 4.495 \cong 4.50$$

The median monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is approximately 4.50 inches.

Mode: The highest frequency does not exist in this sample size. There is no mode.

Midrange: I shall average the dataset from minimum

$$Midrange = \frac{Min + Max}{2} = \frac{0.47 + 17.34}{2} = 8.905 \cong 8.91$$

The midrange monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is approximately 8.91 inches.

Measures of Variability:

These are Range, Variance, Standard Deviation, and Interquartile.

Range:

$$Range = Max - Min = 17.34 - 0.47 = 16.87$$

The difference between the maximum value and minimum value of the monthly maximum precipitation for Miami Area, Florida of January, 2024 to December, 2024 is approximately 16.87 inches.

In calculating the variance and standard deviation all intermediate values shall be rounded to five decimal places.

Variance and Standard Deviation

I shall round all intermediate calculation to 5 decimal places as needed.

Precipitation, x	$x - mean$	$(x - mean)^2$
0.89	-4.90665	24.07538
3.03	-2.76667	7.65444
4.78	-1.01667	1.03361
2.78	-3.01667	9.10028
4.21	-1.58667	2.51751
17.34	11.54333	133.24854
5.44	-0.35667	0.12721
12.83	7.03333	49.46778
9.11	3.31333	10.97818
7.5	1.70333	2.90134
0.47	-5.32667	28.37338
1.18	-4.61667	21.31361
$\Sigma(x - mean)^2 = 290.7912667$		

$$Variance = \frac{\Sigma(x - mean)^2}{n-1} = \frac{290.79127}{12-1} = 26.43557\ inches^2$$

$$Standard\ Deviation = \sqrt{26.43557} = 5.141553267\ inches$$

Interquartile Range, IQR

The data set arranged in ascending order is:

$0.47, 0.89, 1.18, 2.78, 3.03, 4.21, 4.78, 5.44, 7.5, 9.11, 12.83, 17.34$

$n = 12$

$$\frac{1}{4} * 12 = 3$$

$3rd\ position = 1.18\ inches$

$4th\ position = 2.78\ inches$

$$Lower\ Quartile, Q_1 = \frac{1.18 + 2.78}{2} = 1.98$$

$$\frac{3}{4} * 12 = 9$$

$9th\ position = 7.5$

$10th\ position = 9.11$

$$Upper\ Quartile, Q_3 = \frac{1.18 + 2.78}{2} = 8.305$$

$IQR = Q_3 - Q_1 = 8.305 - 1.98 = 6.325\ inches$

The variation in 50% of the monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is 6.325 inches.

Measures of Location

I shall use the five-number summary of data.

$Minimun = 0.47$

The minimum value of the monthly maximum precipitation for Miami Area, Florida of January, 2024 to December, 2024 is 0.47 inches.

First quartile, $Q_1 = 1.98$

25% of the monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is below 1.98 inches.

Middle quartile, $Q_2 = 4.495$

50% of the monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is below 4.495 inches.

Upper quartile, $Q_3 = 8.305$

75% of the monthly maximum precipitation for Miami Area, Florida from January, 2024 to December, 2024 is below 8.305 inches.

$Maximum\ = 17.34$

The maximum value of the monthly maximum precipitation for Miami Area, Florida of January, 2024 to December, 2024 is 17.34 inches.

Works Cited (MLA 9 Format)

Chukwuemeka, Samuel Dominic. “Descriptive Statistics Projects.” *Descriptive Statistics*, conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html#studentsProjects. Accessed 20 Nov. 2025.

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